Reprinted from

Proceedings

46 Congreso Internacional de Americanistas
International Congress of Americanists

Amsterdam, Netherlands 1988



New Directions in American Archaeoastronomy

edited by

Anthony F. Aveni

BAR International Series 454
1988

B.A.R.

5, Centremead, Osney Mead, Oxford OX2 0DQ, England.

GENERAL EDITORS

A.R. Hands, B.Sc., M.A., D.Phil.
D.R. Walker, M.A.

BAR -S454, 1988: 'New Directions in American Archaeoastronomy'

Price £13.00 post free throughout the world. Payments made in dollars must be calculated at the current rate of exchange and $8.00 added to cover exchange charges. Cheques should be made payable to B.A.R. and sent to the above address.

© The Individual Authors, 1988

ISBN 0 86054 583 0

For details of all new B.A.R. publications in print please write to the above address. Information on new titles is sent regularly on request, with no obligation to purchase.

Volumes are distributed from the publisher. All B.A.R. prices are inclusive of postage by surface mail anywhere in the world.

Printed in Great Britain

Astronomical Images and Orientations in the Architecture of Chichen Itza

Susan Milbrath

Chichen Itza is one of the most important Mesoamerican sites for study of astronomical orientations in architecture. The Caracol, with its unusual tower windows, has long been recognized as a Yucatec Maya observatory (Aveni, Gibbs and Hartung 1975). Many other buildings at the site have very interesting astronomical orientations that have not been previously recognized. The most important orientations seem to be towards the zenith sunset, the setting of the Pleiades, and the first appearance of Venus as the evening star at the beginning of the rainy season.[1]

Although most previous work on Mesoamerican architectural orientations has concentrated on measurements and corresponding celestial events, it seems that the alignments may also hold keys to understanding the symbolism of individual buildings. At Chichen Itza the astronomical orientations of various buildings do seem to provide important information for the interpretation of murals and relief sculptures.

Site Chronology

The chronology of Chichen Itza is still much debated because there are two disparate styles of architecture at the site: the Puuc or Chichen-Maya style and the Toltec-Maya style. The Puuc style is named for the hilly region of northern Yucatan, where mosaic architectural decoration predominated during the late Classic period (A.D. 600-900). Chichen-Maya style buildings bear calender inscriptions that cover a period of less than 40 years, between 10.2.0.0.0 and 10.4.0.0.0 (Kelley 1982:Table 1). Kowalski (1987:50) notes that the majority of these dates fall in a 12 year period, from A.D. 869-881. According to Tozzer (1957), the Chichen-Maya or Yucatan-Maya period dates around A.D. 700-950 and the Toltec-Maya period dates between A.D. 987-1263, reflecting an Early Postclassic instrusion from the Central Mexican site of Tula. Kubler (1961) suggests that elements of the Toltec style are actually more developed at Chichen Itza than at Tula, indicating Chichen Itza was probably the originator of certain architectural features at Tula.

A number of scholars have argued that the Great Ball Court complex, with its "Toltec" style features, actually correlates with Late Classic Mesoamerican art styles (Parsons 1969, Cohodas 1978, Pasztory 1972). The chronological placement of the Puuc or Chichen-Maya style as earlier than the Chichen-Toltec style is also being questioned, and now some scholars suggest that both styles existed contemporaneously (Cohodas

1978:Chart III; Wren 1986a, 1986b, Lincoln 1986).

Kowalski (1987:238) suggests a foreign group of Toltecs or Mexicanized Chontal Maya came to Chichen Itza, occupying the Itza city in order to establish authority over trading routes that were in disarray due to political rivalries in northern Yucatan. A relalatively small group of elite outsiders could have inspired a new style of architecture blending foreign elements.

The lack of extensive scientific excavations has hampered accurate dating at Chichen Itza. At present there are only a few clear chronological markers, including radiocarbon dates of A.D. 600 ± 70 and A.D. 780± 70 for the Puuc style Iglesia, and dates of A.D. 790 ± 70 and 810 ± 100 from the Chichen-Toltec style Castillo (Andrews and Andrews 1980:Table 4). The architectural inscriptions range between 10.1.17.15.3 and 10.3.5.5.9, A.D. 867-894 in the Thompson correlation, and the latest date in architectural context is 10.8.10.11.0 (A.D. 998), found at the High Priest's Grave (Krochock 1988:152-153, Fig. 19), a structure that may late in the architectural chronology of Chichen Itza (Wren 1986a; Parsons 1969:Table 7).

Wren's (1986a, 1986b) study of Chichen Itza inscriptions and relief sculpture, and her analysis of a circular stone relief suggest the Great Ball Court Complex dates between A.D. 869-889. Wren notes the circular stone, which may have been originally located in the Great Ball Court, has a Maya style Calendar Round date encircling relief figures that resemble murals and reliefs of the Great Ball Court Complex.

A ninth century date for the Great Ball court seems likely, based on the radiocarbon dates for the Castillo and the span of dates from inscriptions. Fortunately, establishing the exact chronology of the buildings is not essential to our analysis. Our discussion of astronomical alignments is based primarily on repeating celestial cycles, such as the solar zenith sunset and the setting of the Pleiades.

Solar Zenith in Yucatan

In Mesoamerica, as in other areas of the tropics, the sun reaches an overhead or true zenith position at noon twice a year (Aveni 1980:66, Table 3). Unlike temperate zones, where there are four distinct seasons correlating with the equinoxes and solstices, in Mesoamerica there are two seasons: the dry season and rainy season. About 80 percent of the annual rain falls between May and October (Vivo Escoto 1964:201). Throughout much of Mesoamerica the first solar zenith marks the transition from the dry season to the rainy season, when planting takes place today, as it did in ancient times (Milbrath 1981: 270, 281).

The solar zenith date varies with latitude. In the area of Chichen Itza the first solar zenith falls on May 25th, and the second solar zenith on July 20th. In

Yucatan the planting begins in May, usually after the rains begin (Trujillo 1946:203). As a marker for both seasonal change and agricultural activity, the first solar zenith may be the single most important solar event in the Yucatec Maya calendar. In ancient times the second solar zenith also had great importance as the beginning date of the ceremonial calendar - the haab (Aveni 1980:42-43; Tozzer 1941:151-153).

The zenith position has special significance because the sun seems to be closest to the observer, in the "center" as it were. The Maya, with their structured calendar and long distance trade patterns, were aware that the sun reached zenith at varying times of year in different latitudes as they traveled north and south. It is probable that each Maya site was distinguished by its own special ceremony, when the sun reached zenith at an individual local. The overhead position of the sun could be easily determined by watching the changing length of the noontime shadows, with the zenith being the day the sun casts no shadow at noon. Marking the sun's horizon position from a fixed observation point on that date would allow the priests to regulate the agricultural calendar.

Zenith Orientations at Chichen Itza

With a completely flat horizon, the zenith sunset at the latitude of Chichen Itza (at 20° 40′ latitude north) is located at 22° north of west, correlating with an azimuth of 292° (Aveni 1980:Table 9, 1972; Table 1a). An azimuth is our modern measurement incorporating the concept of a 360° circle, starting with the zero point at true astronomical north. Ancient Mesoamerican astronomers did not use this kind of reckoning, and they determined alignments using horizon features and simple sighting devices like crossed sticks (Aveni 1980:Fig. 26). Once the alignments had been established they were often codified in architectural orientations.

Owing to the flat horizon in the area of Chichen Itza, fixing specific horizon points for observation was especially difficult. Given the featureless horizon and the lack of precision instruments for astronomical alignments, it may be that architectural orientations were sometimes off by a degree or even two. Nevertheless, precise orientation is evident in the Caracol, a building designed specifically as an observatory (Aveni, Gibbs and Hartung 1975).

The solar zenith may have been especially important in the architectural orientations at Chichen Itza. The Caracol observatory has a stairway niche with one wall precisely oriented to the zenith sunset at an azimuth of 292° or 22° north of west and the other wall aligned to the maximum northern setting position of Venus (Aveni 1980:Fig. 90). The Castillo, a building with special calendrical significance, has a western orientation of 291°, only one degree from the solar zenith sunset (Fig. 1; Aveni 1980:Table 9, Appendix A).

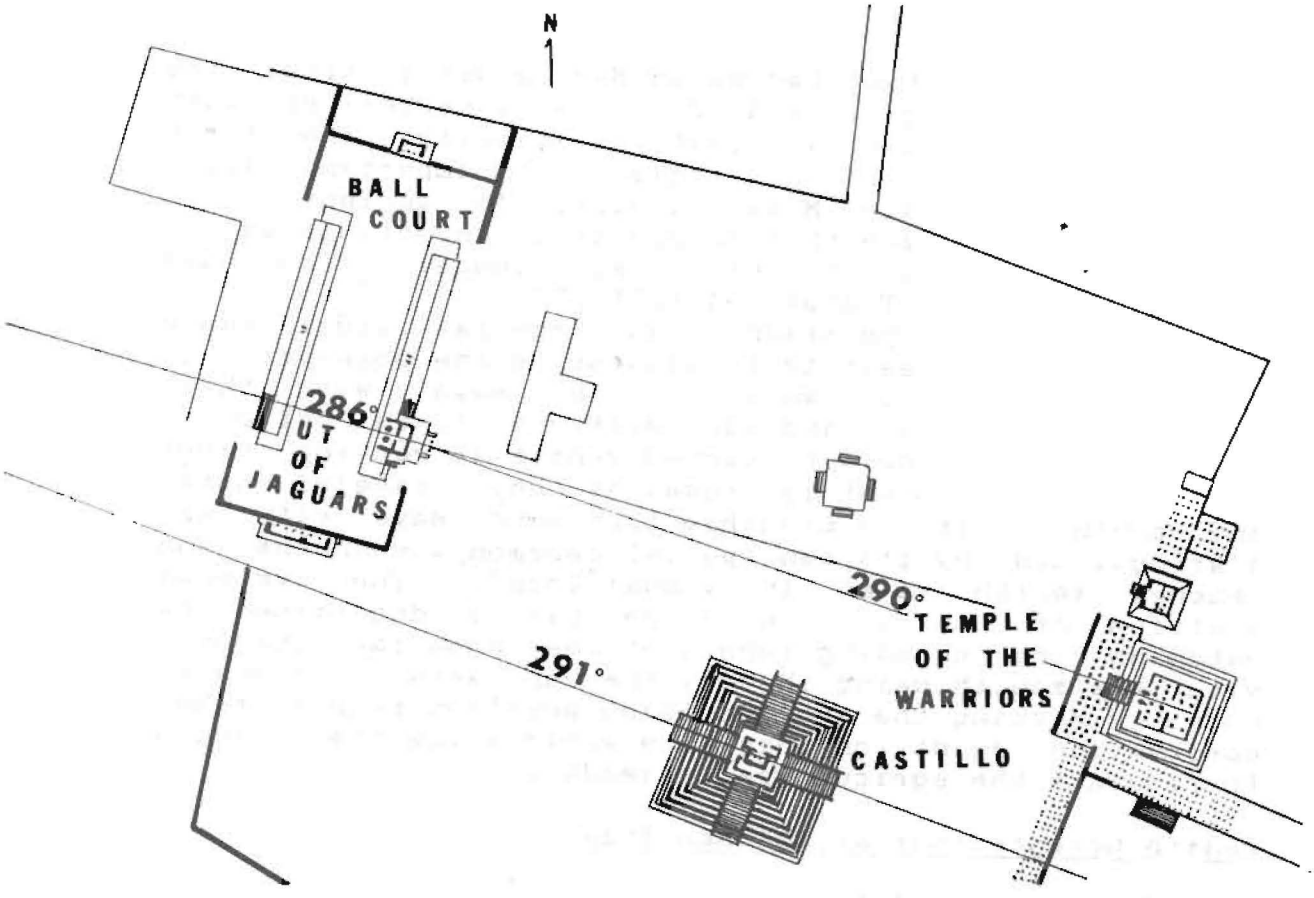


Fig. 1. Map of the astronomical alignments of the North Terrace Group of Chichen Itza, Yucatan, Mexico.

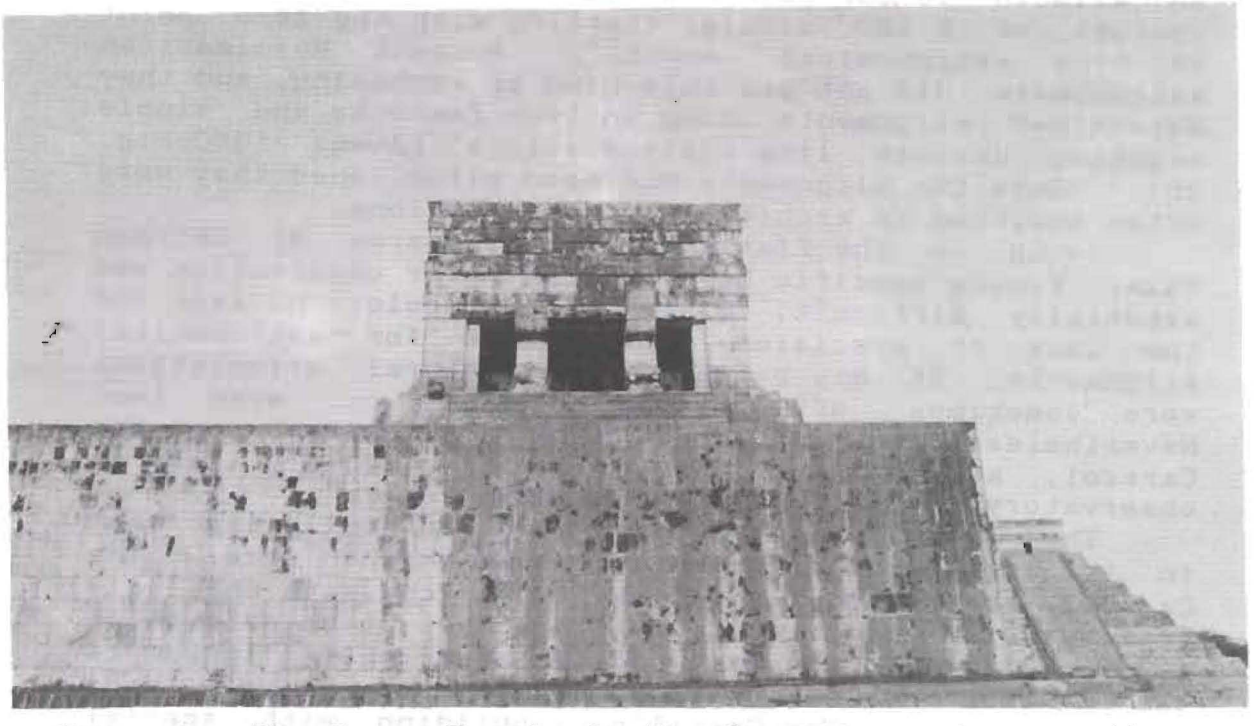


Fig. 2. The Upper Temple fo the Jaguars overlooking the Great Ball Court. The Castillo is in the background.

The Castillo (Fig. 2, background) has exactly 365 steps in the four radiating staircases and temple platform. The pyramid has been the source of great interest because the unusual play of sunlight and shadow at sunset on the equinox creates a pattern along the serpent balastrade that mimics the markings of a diamond-back rattler (Rivard 1969; Krupp 1983:298-299). Coggins (1986) believes the pyramid was constructed to commemorate the completion of the 10th baktun on March 13, 830, a date close to the spring equinox.

Astronomical alignments of the Castillo reflect the fundamental division of the Yucatec Maya year into two seasons or halfs. The west face with its orientation towards the setting sun at the solar zenith (Fig. 1), reflects the dividing point between the dry season and the rainy season. The solar zenith in late May is when the rains begin, which signals the first planting today, as it did in ancient times in Yucatan (Trujillo 1946:203; Milbrath 1981:281).

The east facade of the Castillo, with an orientation of 111° (180° rotation from the west face; Fig. 1), faces the rising sun six months after the solar zenith. This is the time of the solar nadir (November 22) at the latitude of Chichen Itza. The eastern orientation, facing sunrise on the day of the solar nadir, marks the position of the sun at the transition from the rainy season to the dry season in late November.

November is the month the dry season and the maize harvest begins in Yucatan (Vivo Escoto 1964:201; Perez Toro 1942; Trujillo 1946:203). The Mesoamerican maize harvest is a slow process because many of the doubled-over maize ears are allowed to dry on the stalks, and there are two maize crops to harvest, the early maize planted in May and the late maize planted in June. In January the late maize is harvested and work is begun on clearing the fields in preparation for burning (Trujillo 1946:203). January is also the month of the second solar nadir (January 21; Milbrath 1981:279-280). Hence the long maize harvest is bracketed by the two solar nadirs.

The Solar Nadir

Like the solar zenith, the solar nadir is a phenomenon confined to the tropics. The solar nadir is the time when the ecliptic, the path of the sun and planets, stands upright at midnight like a giant wheel with the zodiacal constellations arching overhead. At this time the sun is 180° away, directly below the earth at the latitude of the observer in the tropics.

The solar nadir can be calculated by watching the stars moving opposite to the sun. During the Terminal Classic and Postclassic periods, in the latitudes of the Valley of Mexico and northern Yucatan (which are located

approximately at parallel latitudes), the midnight zenith of the Pleiades correponded approximately to the time of the November solar nadir, when the sun was in opposition to the Pleiades located 180° away. At this time the sun crossed the Milky Way in the region of Scorpius, at the intersection of the Milky Way and the ecliptic. This cosmic crossroads in the sky mirrors the other intersection point of the Milky Way and the ecliptic in the region vicinity of the Pleiades (Milbrath 1980; 1981: 281).

According to Friar Sahagun, every fifty-two years the the Aztecs celebrated the New Fire Ceremony as a sign of world renewal. Sahagun (1950-1982:IV, 143) informs us that the ceremony was timed by the midnight zenith of the Pleiades in November. More recently, scholars have recognized that the ceremony also celebrated the solar nadir (November 30) at the latitude of the Aztec captical of Tenochtitlan (Milbrath 1980; Krupp 1981). We know the Yucatec Maya told time by observations of the Pleiades, and they must also have been aware of the opposition of the sun and Pleiades at the time of the November solar nadir (Milbrath 1980, 1981).

Although we have no direct proof that the solar nadir was recognized among the Yucatec Maya, the architectural orientation of the east facade of the Castillo provides evidence for observation of this solar event. Solar ceremonies may have been performed in the eastern chamber of the Castillo at the time of the solar nadir, when the sunlight at dawn penetrated the dark recesses of the temple.

Lower Temple of the Jaguars

The Lower Temple of the Jaguars has an orientation of 106°, close to the direction of the rising sun at the time of the solar nadir (Fig. 1). The actual orientation is towards sunrise several weeks before the solar nadir in late November and several weeks after the solar nadir in late January. Elaborately carved scenes on the interior of the temple receive the most amount of direct sunlight at sunrise in early November and early February.

Painted reliefs in the interior temple depict five rows of warriors and deity figures, and a prominent sun disk in the uppermost register (Cohodas 1978:220-225, Figs. 16-18; Marquina 1964:Fotos 435, 436). The building may be equated with the Aztec Tlacochcalco, which functioned as an armory and military headquarters (Wren 1986b). Warrior cults and solar images depicted in the upper register indicate a military cult linked to sun worship. The orientation suggests the reliefs represent militaristic ceremonies performed during the dry season, when the rising sun would penetrate the temple and bring the warriors to life.

Both annual solar nadirs coincide with the dry season, bracketing a period of less intense agricultural

activity in Yucatan when warrior cults may have been emphasized. Early Yucatec Maya ethnographic sources note that warfare was usually confined to the dry season (Reed 1964:99). Similarly, ethnohistorical documents inform us that the Aztecs timed all their major military campaigns to coincide with the dry season because roads were most easily traversed at that time (Hassig 1988).



Fig. 3. Feathered serpent columns in the Upper Temple of the Jaguars.

Upper Temple of the Jaguars

Looming over the largest ball court in Mesoamerica, the Upper Temple of the Jaguars has a clear view of the flat horizon (Fig. 2). As in the Temple of the Warriors, entry to the interior space is through twin feathered serpent columns (Figs. 3, 4). Because the

inner temple has been preserved intact, the architectural alignment today remains as it was in the past. The temple has an orientation of 286° (16° north of west), slightly skewed from the actual zenith sunset.



Fig. 4. The solar zenith sunset illuminates the interior of the Upper Temple of the Jaguars.

The positioning of the Upper Temple of the Jaguars creates an interesting play of light and shadow around the time of the solar zenith sunset. At sunset a rectangle of sunlight penetrates the inner chamber like a spotlight passing in between the feathered serpent columns, lighting the jambs and back wall of the inner chamber (Fig. 5). The inner temple was painted with colorful murals that have all but disappeared. Fortunately, they were recorded in the 19th century by Adela Breton (Coggins 1984:Figs. 17-20). The spotlit murals would have added dramatic effects to rituals performed at sunset in the Upper Temple of the Jaguars around the time of the solar zenith.

Prior to modern tree growth atop the west side of the ball court, the last rays of the setting solar zenith sun would have illuminated the carved wooden lintel of the inner temple. On both the outer and inner faces, the lintel depicts a feathered rattlesnake and a sun disk flanking a water-lily god, but only the inner

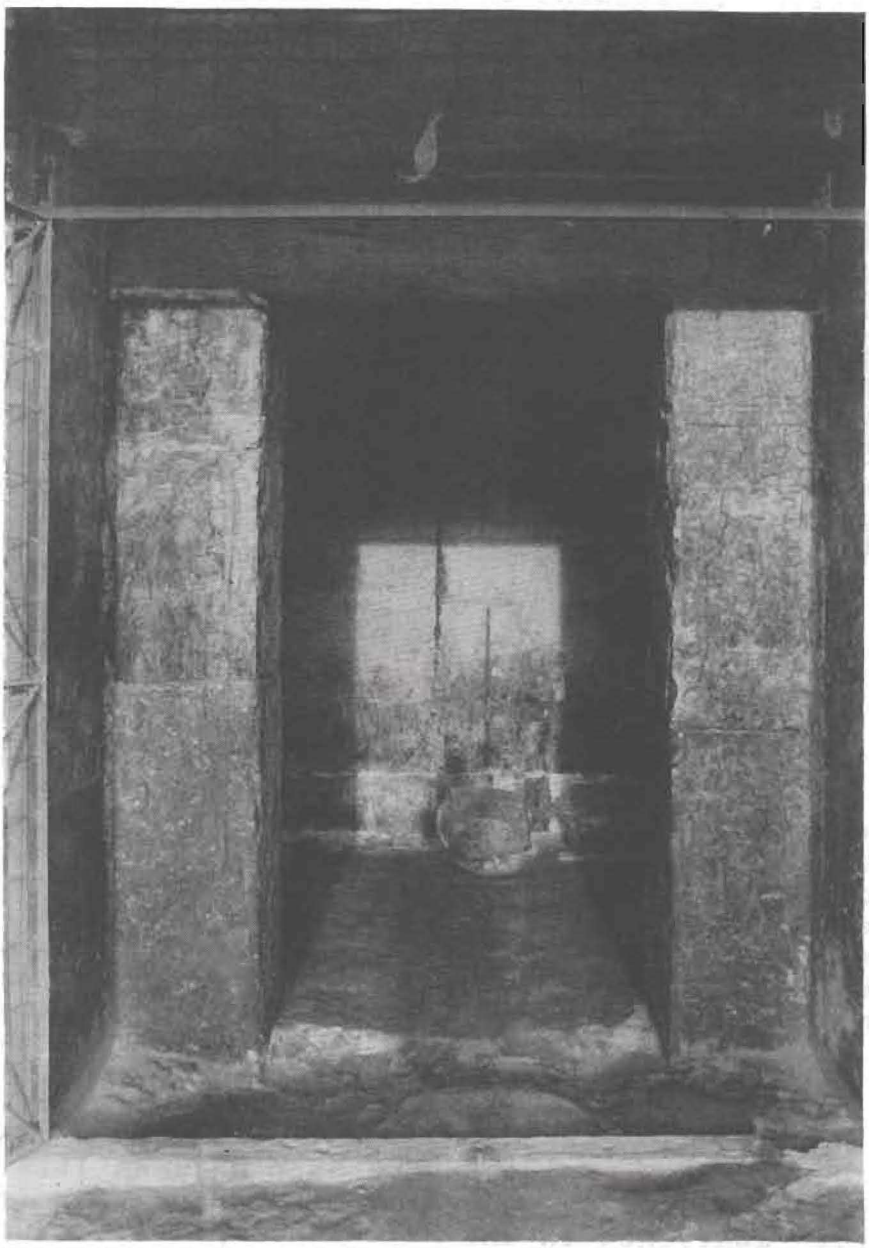


Fig. 5. Light penetrates the inner temple, lighting jambs and back wall.

face has a Venus glyph positioned over the water-lily (Maudslay 1889-1902, v. 3, Pl. 35; Cohodas 1978:Fig. 39). Both faces show two human figures, one with the feathered serpent and the other seated in the sun disk. These figures have been identified as Captain Serpent and Captain Sun Disk (Miller 1977; Coggins 1984:163). More recently Lincoln (1988) proposes these figures are Kukulcan (Sprouting Serpent) and Kakupacak (Fire

Shield), dual rulers of Chichen Itza. Even though they may be historical figures, the prominence of the sun disk and the Mesoamerican Venus god, the feathered serpent (Aveni 1986), indicates a link with the celestial realm. A similar blend of astronomy, history and politics is evident in Aztec monumental sculpture (Milbrath 1983).

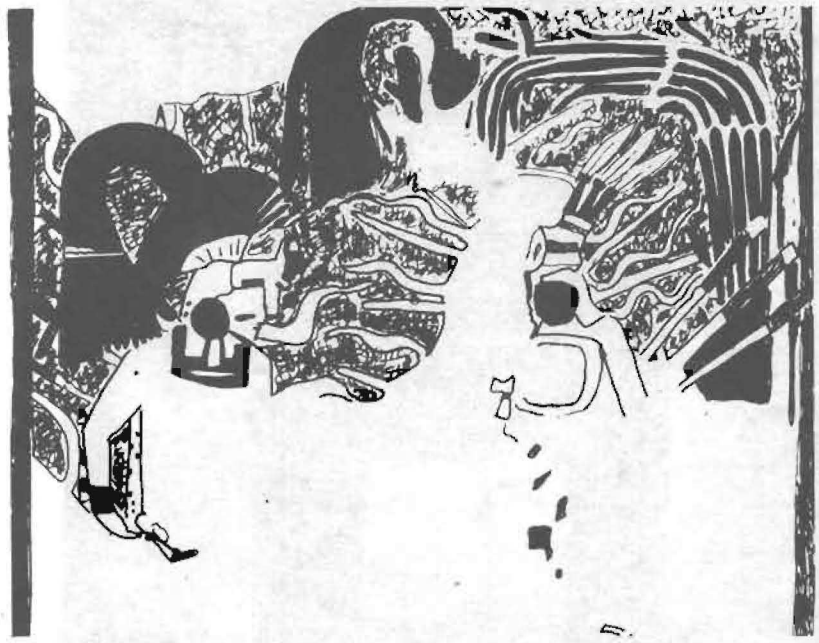


Fig. 6. Mural on rear wall of the inner temple.

When the murals in the Upper Temple of the Jaguars were still intact, two deified rulers in the center panel of the east wall were illuminated around the time of the solar zenith (Fig. 6). According to current interpretations, the figure on the left is Captain Sun Disk, a Maya ruler associated with the sun disk, and the right hand figure is Captain Serpent, a Mexican or Toltec ruler affiliated with the feathered serpent (Miller 1977; Coggins 1984: 160). Astronomy and history may be fused if the figures are in fact co-rulers who claimed celestial patronage or heavenly ancestors, perhaps even the legendary founders of Chichen Itza.

Kelley (1982:2-5) suggests Kakupacal, a person frequently mentioned in the Maya inscriptions at Chichen Itza, is the Itza captain of the chronicles and the founder of Chichen Itza. He believes the murals of the Upper Temple of the Jaguars show a Toltec group conquering the Itza, creators of the Puuc architecture at Chichen Itza. He identifies Quetzalcoatl and Mixcoatl, the leaders of the Toltec army, but he makes no specific identification of the conquered Itza lord in the mural scenes. Miller (1977) suggests the conquered lord is Captain Sun Disk, whom Lincoln (1988) identifies with Kakupacal. It seems more than a coincidence that this ruler's name could be read as a name for the sun itself. We may have another example of a divine "sun

king" among the Maya (Milbrath 1983).

The dynamics of politics and astronomy are developed further in a complex sequence of battle scenes. Coggins (1984) agrees with Miller's (1977) historical interpretation, but adds that Venus and solar positions are recorded in the murals. The scenes of combat in the Upper Temple of the Jaguars probably reflect a link between Venus, astrology and warfare also found in the Classic Maya murals of Bonampak (Lounsbury 1982; Schele and Miller 1986:216-217, Fig. V.6; Coggins 1984:157). The emphasis on astronomical symbolism in the architecture does not invalidate current interpretations relating to historical events (Lincoln 1988; Kelley 1982; Wren 1986a, 1986b), but rather it adds a rich, new dimension. Like the Classic Maya, the builders of Chichen Itza chose to link astronomy and history, most probably to set historical events within a grand scheme related to cosmic events.

The murals in the Upper Temple of the Jaguars may relate to the Venus cycle. Coggins argues that the murals represent both the division of the day into eight parts and an eight-part division of the 584 day Venus cycle, starting with Venus as the morning star before dawn, and following through to the end of the sequence in the middle of the night with Venus in inferior conjunction. The daily motions of the sun are linked conceptually with the 584 day Venus cycle, according to her interpretation. Her analysis combines two unrelated cycles and does not represent a real astronomical situation. Coggin's identification of an eight part mural sequence, however, does provide an intriguing correlation with the Venus cycle. The 365 day year and the 584 day Venus cycle coordinate in such a way that every eight years Venus returns to the same position at the same time of year. We know the Postclassic Yucatec Maya recorded the eight year Venus cycle in the Dresden Codex (Aveni 1980:186). This cycle also may be depicted in the eight mural scenes of the Upper Temple of the Jaguars, with each scene showing Venus in a different position in the sky. The exact patterns that emerge over an eight year period will be discussed below.

The Pleiades and Venus

If observations of Venus were made over the course of eight years, the astronomers may have selected a specific phase of Venus, observing that phase in relation to a specific season, horizon position or stellar coordinate in order to measure Venus positions. The first appearance of Venus as the evening star recurs once every 584 days; but the first appearance of Venus as the evening star during a specific season would occur only once every eight years. Similarly, rising and setting positions of Venus would repeat in a pattern every eight years (Aveni 1986). Interest in the extreme northern setting position of Venus in relation to the solar zenith sunset is evident in the orientation of the

Caracol stairway (Aveni 1980:Fig. 90).

If a stellar coordinate were selected, Venus would be positioned close to a specific star about once a year, but only every eight years would Venus be in a specific phase as it approached the stellar position. The most likely candidate for a stellar coordinate is the Pleiades, which were observed to tell time in Postclassic Yucatan (Tozzer 1942:133).

The Pleiades were called tzab or "the rattlesnake's rattle" by the Yucatec Maya at the time of the conquest (Tozzer 1941:133). The primary language of the residents of Chichen Itza probably was Yucatec Maya (Edmonson personal communication, 1988), which provides a linguistic basis for interpreting the rattlesnake capitals on the serpent columns at Chichen Itza as representations of the Pleiades. This interpretation is reinforced by the fact that the Upper Temple of the Jaguars is orientated towards the position of the Pleiades on the western horizon. The temple's alignment is almost identical to that of Window 1 of the Caracol, which had a midline facing sunset on April 28, with the right side of the window framing the Pleiades setting as they disappeared for a period of a month on that date (Aveni 1980:Fig. 93; Aveni Gibbs and Hartung:Table 1, Fig. 10).[2] Similarly, the wide door of the Upper Temple of the Jaguars framed the Pleiades as they set one month before the solar zenith on May 25th.

Pleiades orientations are evident in other structures with feathered rattlesnake columns, such as the Temple of the Warriors (Fig. 1) and the Temple of the Wall Panels (Retablos; Aveni 1980:Appendix A; Morris et al. 1931; Ruppert 1931). Both of these building have images of the feathered serpent repeated in relief sculptured panels, and they are both aligned towards the setting position of the Pleiades at 290° (with a 0° horizon around A.D. 1000; Aveni 1980:Table 9). This orientation is also close to the position of the zenith sunset (292° with a 0° horizon Aveni 1980: Table 9), but, given the emphasis on the feathered serpent, solar alignment in the architecture may be secondary to the observations of Venus and the Pleiades discussed below.

The Pleiades were probably especially important in the Late Classic and Early Postclassic period because in northern Yucatan they rose at dawn around the time of the solar zenith in late May, after about a month of invisibility. By the Late Postclassic period their setting position also marked the horizon position of the zenith sunset. The Pleiades setting azimuth at 21° latitude north was 289° around A.D. 500. As the end of the Classic period approached, the Pleiades moved closer and closer to the position of the zenith sunset. Around A.D. 1300 they set at 292°, the precise position of the zenith sunset in northern Yucatan (Aveni 1980:Table 9).

The feathered serpent columns at Chichen Itza may relate to an image of Venus as the evening star in conjunction with the Pleiades. The plumed serpent is a widely recognized image of Quetzalcoatl, the god of the

planet Venus in Mesoamerican iconography (Piña Chan 1977; Aveni 1986). An image of Venus wearing the Pleiades, the "rattlesnakes's rattle," might refer to a calendric link between the two astronomical entities.

Looking through astronomical tables compiled by Tuckerman (1964) and comparing the longitudes of Venus with the position of the Pleiades in the Classic and Postclassic periods, there is a very interesting and previously unrecognized correlation between the conjunction of Venus with the Pleiades and the onset of the rainy season in Mesoamerica. Venus moved into conjunction with the Pleiades in a repeating pattern that correlated with the beginning of the rainy season. In its circuit through the background of stars, Venus moved to the position of the Pleiades once a year, and invariably this occurred between mid March and mid June - in other words - around the beginning of the rainy season (Table 1).

Since the Pleiades disappeared in conjunction with the sun near the end of April during the Late Classic and Early Postclassic periods (Aveni 1980:Table 10), the evening star in conjunction with the Pleiades would be visible only in late March through late April, just prior to the onset of the rainy season. And the morning star in conjunction with the Pleiades would be visible only in late May through early June, at the beginning of the rainy season. In either morning or evening star phase, Venus near the Pleiades symbolized the rainy season.

Tracing the dates of the conjunction of Venus with the Pleiades in relation to dates of the first and last appearance of the evening star between A.D. 869-889, the suggested dates for construction of the Great Ball Court Complex (Wren 1986a, 1986b), indicates there is an eight year pattern in relation to the specific dates on which the evening star appears or disappears (Table 2). Table 2 also details the Venus phases in relation to conjunction with the Pleiades, showing that, during the 20 year period studied, Venus was in inferior conjunction near the Pleiades only on two occasions, in the years A.D. 877 and 885. This indicates the eight year pattern was not precise, especially in relation to the short inferior conjunction phase.[3]

A most interesting pattern is evident at eight year intervals in A.D. 869, 877 and 885, when Venus remained very close to the Pleiades around the time of their conjunction with the sun (April 20-May 23, A.D. 1000; Aveni 1980:Table 10) at the beginning of the rainy season. In A.D. 877 this was especially dramatic because, due to retrograde motion, Venus was in the region of the Pleiades for almost three months, and during this period it made the transition through three phases, from the evening star, to its disappearance in inferior conjunction, and reappearance as the morning star.

The years A.D. 870, 878, and 886 also provide an interesting pattern. During these years Venus first

appeared as the evening star crowned by the Pleiades around the spring equinox (March 21). As the evening star rose higher the Pleiades descended closer to the horizon each night, crossing the path of the evening star on April 12 and disappearing in late April, around a month before the solar zenith and the onset of the rainy season. Venus continued its ascent during the rainy season, until it reversed course and descended to make its last appearance as the evening star around the time of the solar nadir at the beginning of the dry season. The Castillo, with its precise orientation to the solar nadir, and the Lower Temple of the Jaguars, with its approximate orientation to the dry season sunrise, may have been used to observe Venus every eight years in its first appearance as the evening star at the beginning of the dry season.

These recurring celestial events may have been a focus for the orientation of the Upper Temple of the Jaguars, which is aligned facing 16° north of west. This orientation may not be towards the sun setting one month before the rainy season, but rather to the evening star and the Pleiades just above the horizon in April. The iconography of the Upper Temple of the Jaguars, emphasizing the feathered rattlesnake, may represent Venus in conjunction with the Pleiades as a symbol of the onset of the rainy season.

Pleiades-Venus Orientations in Mesoamerica

Other buildings at Chichen Itza reflect a pattern of imagery and orientation that can be linked to Venus and the Pleiades. The Temple of the Warriors and the Temple of the Wall Panels have feathered serpent columns like the Upper Temple of the Jaguars, and their alignments of 290° may reflect an orientation towards the Pleiades setting and Venus appearing or disappearing as the evening star at the beginning of the rainy season. The claw-footed Venus monster in the panels of the Temple of the Warriors may be images of Venus in inferior conjunction (Cohodas 1978:Fig. 64; Klein 1976:10).

Šprajc (1988) analysizes the rain-maize complex in relation to Venus imagery and building orientations. He found a connection between rainy season imagery and the evening star. He notes that over many millenia Venus extremes visible on the western horizon (maximum northern and southern horizon positions) coincided with the beginning and end of the Mesoamerican rainy season. Only a few architectural alignments reflect orientations to Venus extremes, but all the examples are related to the evening star horizon extremes, rather than the morning star positions (Šprajc 1988). Our discussion above suggests Pleiades alignments also should be considered in relation to the evening star-rain-maize complex.

Copan Temple 22, with its unusual window and Bicephalic Serpent bearing both Venus and solar imagery,

is an obvious candidate for study of orientations. Anaylsis of the window alignment suggest that the Maya observed the evening star as it made its first appearance in the year prior to a great extreme, when Venus reached a declination that exceeded the limits of the solstice sunset. Similarly, in the year following a great extreme Venus appeared in the window just prior to disappearing in inferior conjunction. The pattern of the first and last appearances bracketing a Venus extreme repeated at eight year intervals, and invariably occurred between April 25 and May 3rd, in other words, at the beginning of the rainy season. The orientation of the window also mirrors the baseline created by the alignment of Stela 10 and 12, marking the position of the sunset on April 12 and September 1 (279° azimuth, Aveni 1980:240; Aveni 986; Closs et al. 1984). The April date falls about a month before the onset of the rainy season.

The dates for the disappearance of the Pleiades can be estimated for the mid 8th century, the probable period for construction of Temple 22, based on a 15 latitude location.. The date for the last appearance of the Pleiades was April 9 circa A.D. 500 circa (Aveni 1971) and several days later around A.D. 750, which would seem to suggest a correlation with the April baseline. As Šprajc (personal communication) points out, however, the Pleiades would not have been visible through the window around A.D. 750 because they set behind the local horizon at an azimuth of 289°, whereas the greatest viewing azimuth allowed by a diagonal sight line through the window is about 285°. This indicates the Pleiades were probably not involved in the Copan window orientation. On the other hand, a number of Mesoamerican temples do seem to reflect orientations towards the Pleiades setting.

The Late Classic Temple of the Plumed Serpent at Xochicalco faces an azimuth of 287°, or 17° north of west (Aveni and Gibbs 1976:513). Around A.D. 500 at the latitude of Xochicalco (18° latitude north), the Pleiades set at 288° with a 3° horizon. Given the pattern linking the Pleiades and the evening star to the onset of the rainy season, it seems likely the orientation and iconography of the pyramid relate to Venus and the Pleiades setting at the beginning of the rainy season.

Teotihuacan, the most important Central Mexican site of the Classic period, has a pyramid decorated with plumed rattlesnakes. The Ciudadela has the same 286° alignment found at the Upper Temple of the Jaguars. Located at the heart of Teotihuacan on the principal axis of the site, the pyramid's orientation coincides with the setting position of the Pleiades around A.D. 1 when the city plan was laid out (Dow 1967:328-329, Figs. 2, 6, 7). Teotihuacan may be the origin of this distinctive alignment, which may relate to observations of the Pleiades and Venus on the horizon just prior to

the rainy season. Studying the pattern of the Pleiades conjunction with the evening star before the onset of the rainy season (Tuckerman 1964), we find it is a very ancient phenomenon, dating at least as far back as the founding of Teotihuacan.

Mexican legends say Quetzalcoatl invented the calendar. Aveni (1986) has pointed out a relationship between the 260 day calendar and the periods of visibility of the morning and evening star. Ancient observations of these periods may have influenced the development of the unique 260 day calendar. Venus was probably observed carefully in relation to the background of stars during the periods of visibility as morning and evening star. Observation of Venus in conjunction with the Pleiades dates back to the Early Classic period or earlier. Ancient priest-astronomers probably encoded a pattern linking Venus, the Pleiades and the rainy season in iconography and orientations of Venus temples throughout Mesoamerica.

1. I would like to thank Anthony Aveni, Sylviane Bouchet, Ramon Carrasco, Clemency Coggins, Lawrence Desmond, Keff Karl Kowalski and Virginia Miller for reading and commenting on earlier drafts of this paper.

2. Aveni (1980:266; Table 10) provides two slightly different dates for the last appearance of the Pleiades around A.D. 1000 at the latitude of Chichen Itza. This difference of about a week (April 19 versus April 28) results from calculations based on varying assumptions and atmospheric conditions (Aveni, personal communication). Since precession slowly affects the disappearance dates as well, it is probably not possible to precisely determine the exact day of disappearance.

3. The inferior and superior conjunction phases of Venus are estimated based on Venus being in a celestial longitude of 7° or closer to the sun (as in Lounsbury 1982). These estimates, however, can only be considered to be approximate. As Šprajc (1988) notes, the exact date of first and last appearances of Venus before and after conjunction cannot be reliably reconstructed due a number of complex variables relating to the arcus visionis - the necessary depression of the sun below the horizon.

Table 1
Venus conjunction with the Pleiades
Based on Hawkins (1966) and Tuckerman (1964)

Pleiades position and Venus longitude approximate

Year	Pleiades position	Venus longitude	Venus phase	Julian date	
A.D.670	37	37	IC	April 24,	670
	37	37	MS	June 6,	671
	37	37	ES	March 25,	672
	37	37	MS	May 9,	673
	37	37	ES	March 20,	674
	37	37	MS	June 10,	674
	37	37	ES	April 9,	675
	37	37	MS	May 23,	676
	37	37	ES	March 12,	677
	37	37	IC	April 24,	678
A.D.900	40	40	MS	May 10,	900
	40	40	ES**	April 13,	901
	40	40	MS	June 11,	901
	40	40	ES	April 10,	902
	40	40	MS	May 26,	903
	40	40	ES	March 13,	904
	40	40	SC	April 25,	905
	40	40	MS	June 7,	906
	40	40	ES	March 27,	907
	40	40	MS	May 10,	908
A.D.1500	49	49	ES	April 7,	1500
	49	49	MS	May 22,	1501
	49	49	ES	March 16,	1502
	49	49	ES*	April 23,	1503
	49	49	MS	June 5,	1504
	49	49	ES	March 24,	1505
	49	49	MS+	May 9,	1506
	49	49	MS	June 17,	1507
	49	49	ES	April 7,	1508

MS=morning star
ES=evening star
SC=superior conjunction
IC=inferior conjunction
*=first appearance ES after invisibility in superior conjunction
**=last appearance ES before invisibility in inferior conjunction
++=first appearance MS after invisibility in inferior conjunction
+=last appearance MS before invisibility in superior conjunction

Table 2
Venus in conjunction with the Pleiades
A.D. 869-889

Dates in Julian calendar, add 4 days for Gregorian dates
Pleiades position and Venus longitude approximate

Pleiades with Venus	Pleiades	Venus	Venus phase	first of ES	last of ES
5/9/869	40	40	MS	---------	4/25/869
6/5/869	40	40	MS	same as above	
4/12/870	40	40	ES	3/26/870	11/28/870
5/27/871	40	40	MS	10/22/871	--------
3/15/872	40	40	ES	---------	7/5/872
4/27/873	40	40	SC	5/29/873	--------
6/8/874	40	40	MS	---------	2/10/874
3/29/875	40	40	ES	1/9/875	9/16/875
5/12/876	40	40	MS	8/6/876	--------
3/13/877	40	40	ES	---------	4/23/877
5/2/877	40	40	IC	same as above	
6/8/877	40	40	MS	same as above	
4/12/878	40	40	ES	3/24/878	11/26/878
5/27/879	40	40	MS	10/20/879	--------
3/15/880	40	40	ES	---------	7/1/880
4/27/881	40	40	SC	5/27/881	--------
6/7/882	40	40	MS	---------	2/8/882
3/29/883	40	40	ES	1/7/883	9/9/883
5/12/884	40	40	MS	8/9/884	--------
4/26/885	40	40	IC	---------	4/21/885
5/10/885	40	40	MS	same as above	
4/12/886	40	40	ES	3/22/886	11/24/886
5/27/887	40	40	MS	10/18/887	--------
3/14/888	40	40	ES	---------	6/29/888
4/26/889	40	40	SC	5/30/889	--------

MS=morning star
ES=evening star
SC=superior conjunction
IC=inferior conjunction

Andrews, E. W IV and E. W. Andrews V
1980 Excavations at Dzibilchaltun, Yucatan, Mexico. Tulane University, Middle American Research Institute, Publication 48.

Aveni, A. F.
1971 Tables of rise-set azimuths and heliacal rise-set dates. Colgate University Computer Center, Hamilton, New York.

1972 Astronomical tables intended for use in astro-archaeological studies. American Antiquity 37(4):531-540.

1980 Skywatchers of ancient Mexico. University of Texas Press, Austin.

1986 The real Venus-Kukulcan in the Maya inscriptions and alignments. Paper presented at the Sixth Mesa Redonda de Palenque, Mexico.

Aveni, A. F. and S. L. Gibbs
1976 On the orientation of Precolumbian buildings in Central Mexico. American Antiquity 41(4):510-517.

Aveni, A. F., S. L. Gibbs and H. Hartung
1975 The Caracol Tower at Chichen Itza: An ancient astronomical observatory? Science 188 (4192):977-985.

Closs, M. P., Aveni, A. F. and Crowley, B.
1984 The planet Venus and Temple 22 at Copan. Indiana 9(1):221-247.

Cohodas, M.
1978 The Great Ball Court at Chichen Itza, Yucatan, Mexico. Garland Publishing Co., New York.

Coggins, C.
1983 The Stucco decoration and architectural assemblage of Str. 1-sub, Dzibilchaltun, Yucatan, Mexico. Tulane University, Middle American Research Institute, Publication 49.

1984 "Murals in the Upper Temple of the Jaguars, Chichen Itza." in Cenote of Sacrifice: Maya treasures of the Sacred Well at Chichen Itza. edited by C. Coggins and O. C. Shane, pp. 157-165. University of Texas Press, Austin.

1986 A New Sun at Chichen Itza. Paper presented at the Second Oxford International Conference on Archaeoastronomy. January 1986, Merida, Yucatan, Mexico, submitted for publication.

Dow, J. W.
1967 Astronomical orientations at Teotihuacan, a case study in astro-archaeology. American Antiquity 32(3):326-334.

Hawkins, G. S.
1966 Astro-archaeology. Research in Space Science, Special Report, 226. Smithsonian Institution Astrophysical Observatory, Cambridge, Mass.

Hassig, R.
1988 Structure and development of the Aztec empire. Paper presented at the Fifty-third Annual Meeting of the Society for American Archaeology, Phoenix, Arizona.

Kelley, D. H.
1982 "Notes on Puuc inscriptions and history." Supplement to The Puuc: New Perpectives, edited by L. Mills. Central College, Pella, Iowa.

Klein, C.
1976 The identity of the central deity of the Aztec Calendar Stone. Art Bulletin LVIII(1):1-12

Kowalski, J. K.
1987 The House of the Governor. University of Oklahoma Press, Norman and London.

Krochock, R. J.
1988 The Hieroglyphic Inscriptions and Iconography of Temple of the Four Lintels and Related Monuments, Chichen Itza, Yucatan, Mexico. Masters of Arts Thesis, University of Texas, Austin.

Krupp, E. C.
1982 The "binding of the years," the Pleiades, and the nadir sun. Archaeoastronomy V(1):9-13.

1983 Echos from Ancient Skies. Random House, New York.

Kubler, G.
1961 Chichen Itza y Tula. Estudios de Cultura Maya 1:47-80.

Lincoln, C.
1986 The chronology of Chichen Itza: A review of the literature. In Late Lowland Maya Civilization: From Classic to Postclassic, edited by E.W. Andrews V and J. A. Sabloff, pp. 141-196. Santa Fe: School of American Research. Advanced Seminar Series.

1988 Dual Kingship at Chichen Itza, Yucatan. Paper presented at the Fifty-third Annual Meeting of the Society for American Archaeology, Phoenix, Arizona.

Lounsbury, F.
1982 Astronomical knowledge and its uses at Bonampak, Mexico. In Archaeoastronomy in the New World, edited by A. F. Aveni, pp. 143-166. Cambridge University Press, Cambridge:

Marquina, I.
1964 Arquitectura Prehispanica. INAH, Mexico.

Maudslay, A.P.
1889-1902 Archaeology. Biologia Centrali-Americana (56-60). Vol. 3. Porter, London.

Milbrath, S.
1980 Star gods and astronomy of the Aztecs. In La antropologia americanista en la actualidad, Homenaje a Raphael Girard, Vol. 1. pp. 289-305. Editores Mexicanos Unidos, Mexico.

1981 Astronomical imagery in the serpent sequence of the Codex Madrid. In Archaeoastronomy in the Americas, edited by R. A. Williamson, pp. 263-284. Ballena Press, Los Altos and Center for Archaeoastronomy, College Park.

1983 The "sun king" in ancient America. Paper presented at the Annual Meeting of the College Art Association, Philadelphia, 1983.

Miller, A. G.
1977 Captains of the Itza: Unpublished Mural Evidence from Chichen Itza. In Social Process in Maya Prehistory: Studies in Honour of Sir Eric Thompson, edited by N. Hammond, pp. 197-225. Academic Press, New York.

Morris, E. H., Charlot, J. and A. A. Morris
1931 The Temple of the Warriors. Carneige Institution of Washington, Publication 406. 2 Vols.

Parsons, L. A.
1969 Bilbao, Guatemala, Vol. 2. Publications in Anthropology, no. 12. Milwaukee Public Museum, Milwaukee.

Pasztory, E.
1972 The historical and religous significance of the Middle Classic ball game. In Religion en Mesoamerica. XII Mesa Redonda, pp. 441-455. Sociedad Mexicana de Antropologia, Mexico.

Perez Toro, A.
1942 La Milpa. Merida, Yucatan, Mexico.

Pina Chan, R.
1977 Quetzalcoatl: Serpiente Emplumada., Fondo de Cultura Economica. Mexico.

Reed, N.
1964 The Cast War of Yucatan. Stanford University Press, Stanford.

Ruppert, K.
1931 The Temple of the Wall Panels. Carnegie Institution of Washington, Publication 403, contribution 3, Washington.

Sahagun, B. de
1950-1982 The Florentine Codex. Translated by A. J. O. Anderson and C. E. Dibble. School for American Research, the University of Utah Press, Santa Fe, New Mexico. 13 Vols.

Schele, L. and M. Miller
1986 The Blood of Kings: Dynasty and Ritual in Maya Art. Kimbell Art Museum, Fort Worth.

Sprajc, Ivan
1988 Venus-rain-maize complex in Mesoamerica: associated with the evening star? ms.

Tozzer, A.
1941 Landa's Relacion de las cosas de Yucatan. Papers of the Peabody Museum, Harvard University, Vol. 18. Cambridge, Mass.

1957 Chichen Itza and its Cenote of Sacrifice. Memoirs of the Peabody Museum, Harvard University, Vols. 11-12. Cambridge, Mass.

Trujillo, C. A. E.
1946 Enciclopedia Yucatanense. Vol. VI. Gobierno de Yucatan, Mexico.

Vivo Escoto, J. A.
1964 Weather and climate of Mexico and Central America. In Handbook of Middle American Indians Vol. 1, edited by Robert C. West, pp. 187-215. University of Texas Press, Austin.

Wren, L.
1986a The Great Ball Court Stone from Chichen Itza. ms. Paper presented at the Mesa Redonda de Palenque, 1986.

1986b Elite interaction during the Terminal Classic period of the northern Maya Lowlands: Evidence from the reliefs of the North Temple of the Great Ball Court at Chichen Itza. ms.